

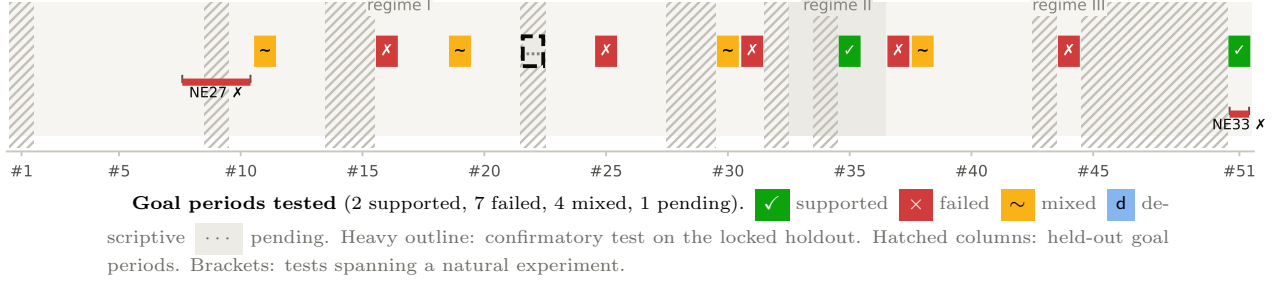
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 0.5/5 Usefulness 2/5

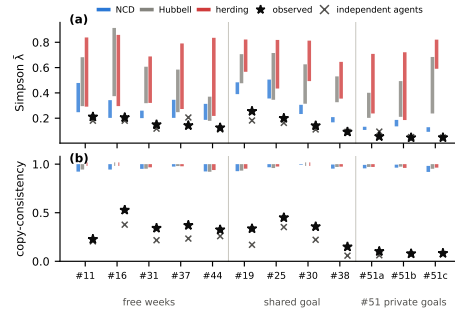
Bottom line. Free LLM agents state private sub-tasks but work on shared repos when one exists; NCD fails, neutral copying fits at best, and copying follows reading.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Each agent slot carries one project label per 30-min window. The label is a cluster of the agent's self-written goal, the repo it links, or (round 1b) the repo it commits to. Exact finite- N simulations with the real label gaps fit three models to 9 statistics: neutral cooperative dynamics (NCD), Hubbell drift and a herding rival. **NCD fails** in free weeks (4/5 failed, 1 mixed). In rounds 1 and 1b the joint check rejects all three models on stated goals in every free week ($p < 0.01$). In free weeks 72–91% of goal clusters have a single agent. A second embedding model and style removal (round 1b) leave this intact. #35 is a control week with a known project: one fork per room. Its stated goals are just as fragmented, but 96–100% of work sits on the room's fork. Goal clusters measure sub-tasks, not projects. Work labels concentrate on a shared repo where one exists. There Hubbell-like copying can fit (#30), without NCD's signature. Copying follows reading: a project named in a read message predicts a switch (OR 8.6 against unread).



Observables. Per period, goal clusters (km24). (a) The observed Simpson λ (star) against the 95% predictive ranges of the three fitted copying models (bars). The cross is an independent-agents null that shifts each agent's day sequence. The observation sits below all three models and close to the null. (b) The share of project switches that join a project another agent holds. Every copying model predicts ≈ 1 . Free and shared weeks give 0.15–0.53, and #51 (private roles) gives 0.08–0.10.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 06 (Piñero et al. 2025). N individuals (agent slots) each hold one species (project). In each step, with probability μ , a new project replaces a random slot C . Otherwise C copies the project of slot A only if A and a third slot B hold different projects. This is cooperation: rare projects recruit more per capita. $\lambda = \sum_i x_i^2$ is the Simpson index of the shares $x_i = n_i/N$. The rivals are Hubbell (always copy) and herding (copy favoured when A and B agree).

$$b_n = (1 - \mu) \frac{n}{N} \left(1 - \frac{n}{N}\right)^2, \quad P_n \propto \frac{(1 - \mu)^n}{n} e^{-(n - N\lambda^*)^2 / 2N}$$

$$\frac{1}{\mu} = \sqrt{\frac{\pi N}{2}} \operatorname{erfcx}\left(\sqrt{\frac{N}{2}} (\mu - \lambda^*)\right), \quad \mu_B = \frac{e^{-2}}{\sqrt{2\pi N}}, \quad \mu_L = \sqrt{\frac{2}{\pi N}}$$

$$P(\text{recruit to } i) \propto x_i e^{\beta x_i} : \quad \beta < 0 \text{ NCD}, \quad 0 \text{ Hubbell}, \quad > 0 \text{ herding}$$

Neutral cooperative dynamics does not describe these swarms. What agents *say* they do is a spread of private sub-tasks in every week, so stated-goal abundances are a poor project census. What agents *work* on concentrates on a shared repo when the task has one. It spreads out when the task gives each agent its own artifact or role (#44 free room, #51), as H11 finds for effort. Copying runs through reading: a read mention raises the switch odds 8.6-fold over an unread one. To get a shared project, assign it or announce it in a message the agents read. To monitor, compute the work-label singleton fraction and the copy-consistency.

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 NCD beats rivals (free)	4 failed, 1 mixed → same count (#11, #16 swap)	failed
Adequacy, stated goals	all 3 models rejected in 5/5 free weeks (both rounds)	failed
Singletons, goals (free)	0.79–0.91 → 0.72–0.84 (gte, style removed)	robust
P2 rare-project advantage	#31 $\hat{\beta}$ $-0.78 \rightarrow +1.47$; sign varies by embedding	failed
Work vs goals (R1b-3a)	work more concentrated: $\bar{\lambda}$ 5/5, supported copy 4/5	
NCD on work (R1b-3b)	inadequate 4/5; #30 NCD and Hubbell adequate, no core	supported
Copying needs reading	OR read vs unread 8.6 (attention), 5.6 (work)	supported
#35 known universe (native)	work 96–100% on own fork; goal singletons 0.76	supported
#51 rivals share (native)	goals 2/3 blocks; attention 0/3; 1/3 work 0/3	
#44 assigned vs free (native)	assigned arm more concentrated in 3 spaces	mostly

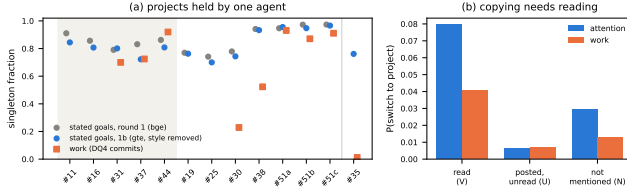
GOAL PERIODS IN DETAIL

G22 ... 1b: pending (confirmatory, not run) (**holdout**)
G11 ~ 1b: mixed (P1 by rule; all models still inadequate)
G16 × 1b: failed (km24 only; all models inadequate)
G19 ~ 1b: mixed (P6 unchanged; all models inadequate)
G25 × 1b: failed (P6: conformist no longer beats NCD)
G30 ~ 1b: mixed (P6); work labels: NCD and Hubbell adequate...
G31 × 1b: failed (P1 6/6; work labels also inadequate; P2 n...
G35 ✓ 1b: supported (native: known universe recovered; goal...
G37 × 1b: failed (km24 only; all models inadequate)
G38 ~ 1b: mixed (P6; km24 all inadequate; work: Hubbell ade...
+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. (a) The share of projects that one agent holds. Stated goals (grey round 1, blue round 1b) stay high everywhere, including #35, where the project is known. Work labels (orange) drop where a shared repo exists (#30, #35, #38). Shaded: free weeks. (b) The probability of a switch to a project that another agent holds. The classes: a message naming it is read, posted but unread, or absent.

CAVEATS

Compute was scarce. The full estimator (all clusterings) reran on #11 and #31 only. Elsewhere round 1b uses the primary clustering, the work labels and model-free statistics. Likelihood ratios near ± 2 and $\hat{\beta}$ depend on the embedding. In two-room weeks the unread class is mostly the other room's messages. The day-shift null cannot separate a persistent shared field (room assignment) from copying. Work labels cover public commits only, dense from #30.

ROUND 2

Fit NCD and Hubbell with agent-specific fields on work labels. Anchor stated goals to repos. Run the #22 confirmation after sign-off, with work labels as a secondary.

No round-2 redirects yet.

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